

Problem Statement and Goals

Trajecto

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Table 1: Revision History

Date	Developer(s)	Change
2026-01-16	Zhuo Zhang	Initial draft
2026-04-15	Zhuo Zhang	Updated problem, input/output, and goals for detector-line and region constraints

1 Problem Statement

1.1 Problem

Trajecto addresses prediction of charged-particle motion in piecewise-uniform electromagnetic regions and detector-line outcomes. The software models a single particle moving in the x-y plane under user-defined electric and magnetic fields, where each field region is rectangular and adjacent to neighboring regions. All regions must have identical width and height, and the union of all regions must form one larger rectangle. The main result is whether the particle hits a user-defined detector line and, if it does, when and where the hit occurs.

1.2 Inputs and Outputs

The inputs are particle properties, detector-line specification, and field-region specification. Particle properties include charge, mass, and initial velocity. Detector-line inputs include orientation (horizontal or vertical), location, and length. Field-region inputs include each region's four-corner coordinates and its electric/magnetic values (E and B). Input validity requires all regions to have equal size, to be placed edge-to-edge, and to form a rectangular union. The outputs are:

- -1 when no detector-line hit occurs within the simulation scope.
- t_{hit} and hit location when a detector-line hit occurs.

1.3 Stakeholders

- Primary use stakeholders: instructors or students who want to confirm or analyze motions of charged-particle in different conditions of fields.
- Project stakeholders: the Drasil maintainers and course instructors who will check the artifacts, traceability and quality of the document.
- Secondary stakeholders: future CAS741 students who wish to reuse, refer or extend the case study.

1.4 Environment

This software runs on personal computer with Ubuntu 22.04 LTS. User interacts via command line. Inputs are provided through a text-based configuration and outputs are given by portable and standard formats in CSV. The documentation and generated artifacts are under Drasil workflow.

2 Goals

- This software will compute the particle trajectory through piecewise-uniform electric and magnetic regions defined by rectangular, equal-sized, adjacent regions.
- This software will determine detector-line outcomes for a detector that is either horizontal or vertical with user-specified location and length.
- This software will report -1 for no-hit cases, and report t_{hit} plus hit location for hit cases.

3 Stretch Goals

- This software will solve for unknown parameter given a target detector location.
- This software will add error estimation to improve accuracy across different parameters.